N Vera Chau

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Nationality: American

Professional Experience

2025-current Visiting Assistant Professor of Finance, S.C. Johnson College of Business & Charles H. Dyson School of Applied Economics and Management, Cornell University

2024-2025 *Parental Leave*

2022-2025 Assistant Professor of Finance, Geneva Finance Research Institute, University of Geneva & Swiss Finance Institute

2012-2015 Research Associate, Paul Milstein Center for Real Estate, Columbia Business School

2010-2012 Analyst, Investment Banking, Rodman & Renshaw (later KLR Group, LLC)
Sector: Oil & Gas

Education

2015 - 2022 PhD in Finance, **Booth School of Business- University of Chicago**

2012-2015 Non-Degree Post-Bacc[1], **Columbia University**

2007-2010 BA in Economics, BA in International Relations, **New York University**[2]
(cum laude)

Spring, 2009 Study Abroad, New York University in Paris

[1]Real Analysis I & II, Graduate Analysis and Probability I & II, Graduate Micro, ODE, Linear Algebra

[2]Dean's List, Presidential Honors Scholar, Highest Honors in Economics

Works in Progress

• Local Networks and Technology Adoption during Industry-Wide Technological Change

Presentations: University of Virginia (McIntire), Università della Svizzera italiana (Lugano), University of Zurich

I study how firms' adoption responses to local network information change during the U.S. hydraulic-fracturing transition. I combine spatial panel models that characterize local network signals with a bunching estimator that isolates peer drilling as a plausibly exogenous source of new information. At the same local-development stage, the adoption response to peer activity is negative in markets that develop when the technology is young industry-wide and positive in markets that develop later. A calibrated model uses the gap between adoption and investment responses to distinguish returns to local evidence from starting costs as channels through which industry development shapes firms' use of local network information.

• Integrated Intermediation & Fintech Market Power
with Greg Buchak & Adam Jørring

Presentations: Olin Finance Conference at WashU, Fixed Income and Financial Institutions (FIFI) Conference (University of South Carolina), University of Geneva Economics dept

We study the disintermediation of loan origination and loan servicing in the U.S. mortgage market. Using plausibly exogenous variation from risk-based capital rule changes, we show that borrowers with vertically integrated originator-servicers are significantly more likely to engage in beneficial refinance than those with stand-alone servicers. They are also more likely to refinance with the servicer's associated lending business. We build and estimate a dynamic structural model with adverse selection to separate the selection of borrowers from a causal effect of integration through cost advantages and borrower convenience. Nearly the entire effect is causal, implying that disintermediated origination and monitoring can have significant downsides for borrowers.

• What Does It Mean to Adopt a Technology?

New technology diffuses when firms incorporate it into production. Adoption can be an overhaul that replaces old technology or a diversification that adds new capital without retiring the old. Using the US shale boom, I develop a heterogeneous firm model in which technology productivity, entry costs and joint operating costs determine whether producers switch to horizontal drilling or straddle both. I estimate the model by simulated method of moments using a neural network solver. Leaders under the old technology need not lead under the new. Switchers generate two thirds of

shared industry learning. Lowering entry costs raises the share of adopters from 52 to 76 percent.

• Off-Balance-Sheet Costs and Credit Supply

Operational costs increasingly shape mortgage lenders' ability to extend credit. Nonbank lenders rely on warehouse funding rather than deposits and the growing complexity of household finance raises the resources required after a loan is made. Credit supply can therefore depend on operational capacity, such as servicing existing loans, rather than only on the capacity to fund new lending. I use plausibly exogenous differences in Fannie Mae and Freddie Mac advancing rules to study advance burdens during the COVID-19 forbearance wave. Lenders more exposed to Fannie Mae's longer obligation reduced originations by about 14 percent per standard deviation of exposure. Less-exposed lenders absorbed much of the displaced lending.

• Technology Adoption Along a Moving Frontier

Technological change, climate tipping risk and dated policy changes unfold along paths that are nonstationary, change regimes or contain calendar breaks. Recursive methods accommodate these environments only after the economist constructs a tailored finite-dimensional state and law of motion. I develop a transition operator that maps exogenous paths into optimal decision sequences, with a recurrent network learning the history summary relevant for decisions. I apply the method to firm investment and technology adoption along a moving frontier. Adoption and investment are dynamically complementary. The same technology gap can support different decisions when observed histories imply acceleration, plateau or reversal.

• Income Histories and the Marginal Propensity to Consume

I extend the permanent income hypothesis from a recursive policy function to an operator mapping income histories and dated environments into consumption plans. The operator handles heterogeneous ARMA dynamics, hidden regimes, long memory and irregular policy calendars. A neural primal solver makes simulated method of moments computationally feasible. An MPC becomes an event-specific derivative that decomposes into the standard annuity response to cash and a news term measuring revisions to expected future income. At the current benchmark calibration, estimated earnings-process heterogeneity generates the steep cross-sectional gradient in responses to ordinary income changes without matching their levels. Predicted windfall responses remain well below empirical estimates.

Teaching Experience

Professor:
Managerial Finance (MBA core), Cornell Johnson

Intro to AI in Business (undergraduate major), Cornell Dyson
Financial Markets & Institutions (undergraduate), Cornell Dyson
Finance (undergraduate business minor), Cornell Dyson
Advanced Asset Pricing (undergraduate), University of Geneva
Financial Markets (undergraduate), University of Geneva

Teaching Assistant:

Financial Econometrics / Time-Series Analysis (MBA), Professor Jeffrey Russell, University of Chicago Booth School of Business
Business Statistics (MBA and EMBA), Professor Jeffrey Russell, University of Chicago Booth School of Business
Citadel Statistics (new engineer summer training), Professor Jeffrey Russell, Citadel
Entrepreneurial Finance and Private Equity, Professor Steven Kaplan, University of Chicago Booth School of Business
Real Estate Finance and Advanced Real Estate Finance (MBA), Professors Tomasz Piskorski and Christopher Mayer, Columbia Business School

Technical Languages

Python, R, Fortran, Matlab, Stata, Eviews

PhD References

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